

Tick Tock: Maintaining Time

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2026-09-10
Montréal



RustConf
2026

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 - parallel API surfaces: panicking and fallible

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- maintain a high bar for quality and correctness
- API evolves, but your build never breaks

Types Overview

- Core

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 - Month
 - Weekday

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- Format description

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- Format description

- Low-level types for formatting & parsing

Helper Macros

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Date

```
date!(2026-09-10)
```

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`date!(2026-09-10)`

Time

`time!(2:47:00 PM)`

Helper Macros

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Not everything belongs in time!

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`powerfmt`

helpers for implementing `Display`

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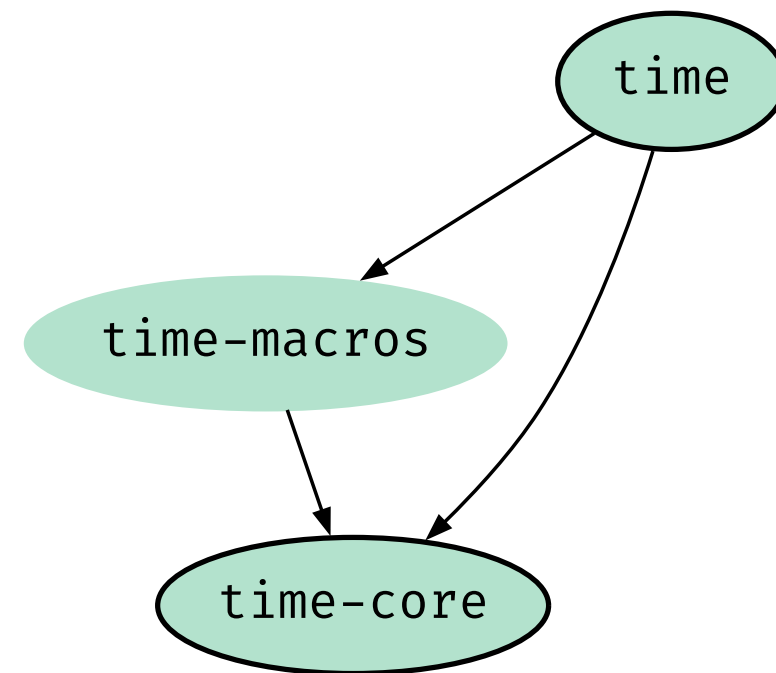
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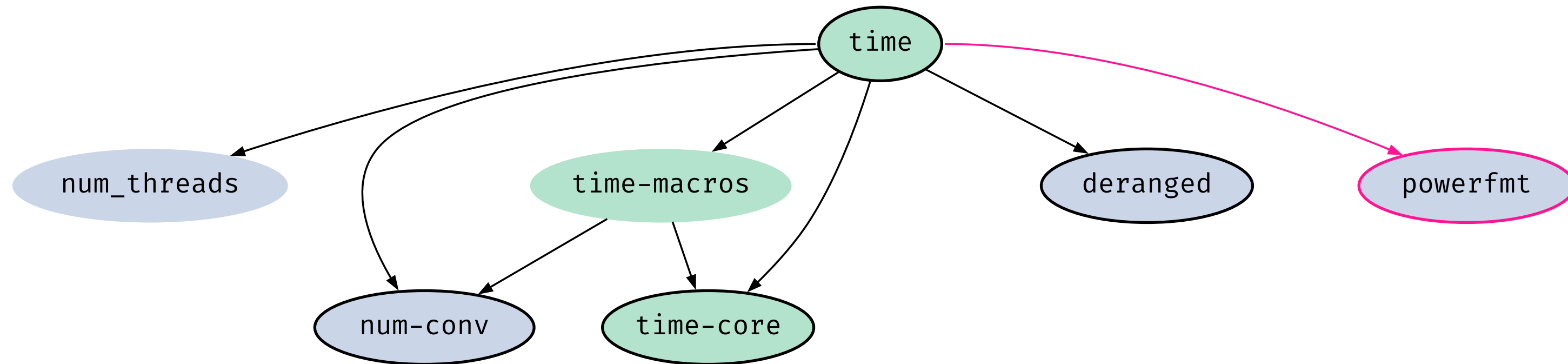
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<code>deranged</code>	ranged integers
<code>zoneinfo</code>	TZif file reading & writing
<code>zoneinfo-static</code>	precompiled time zone database

Dependency Tree



□ mandatory ■ time workspace

Dependency Tree



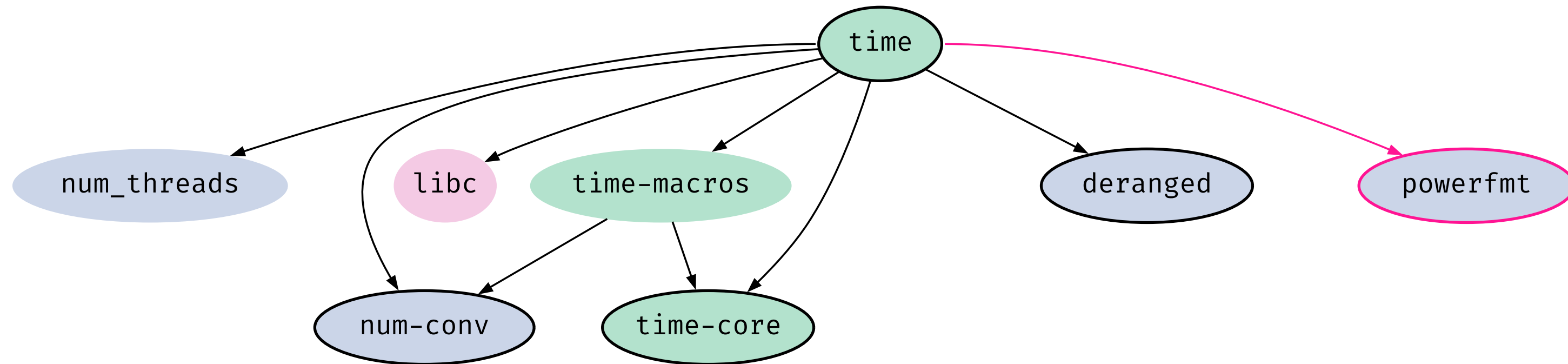
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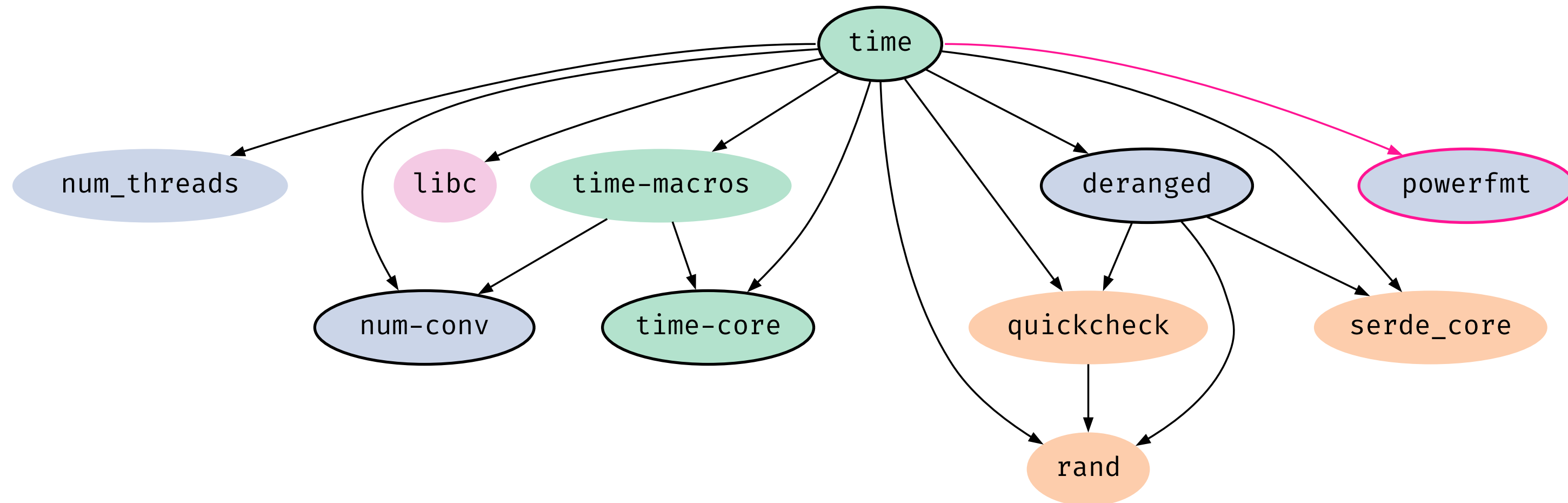
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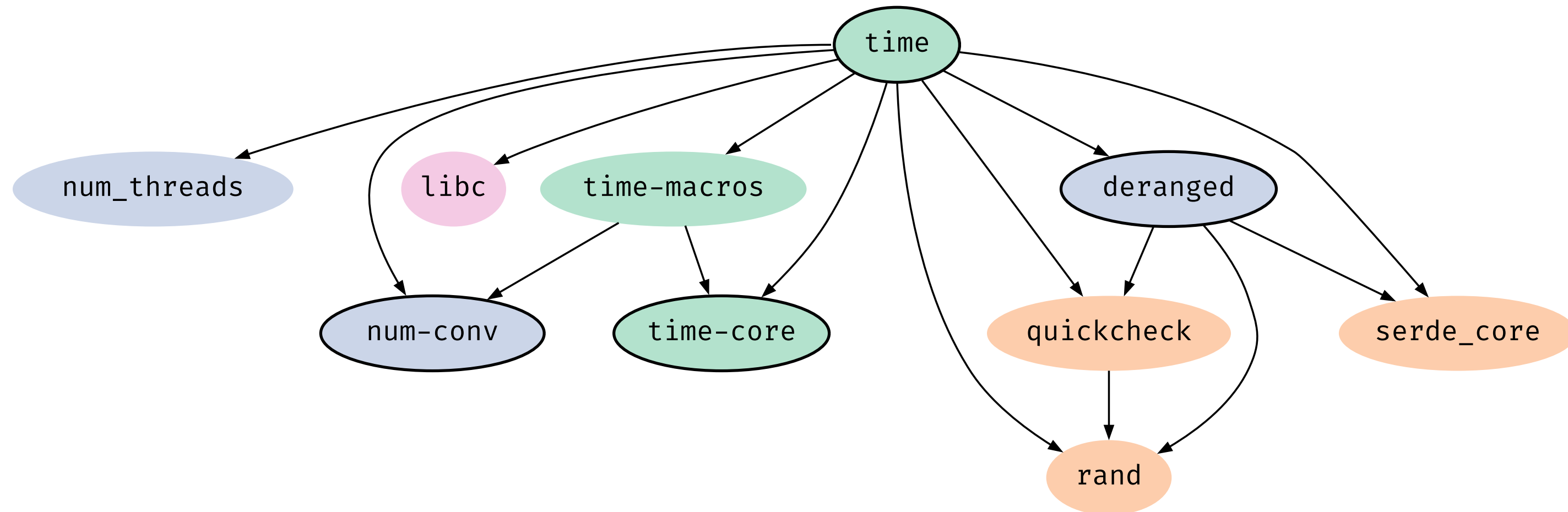
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Dependency Tree

(future possibility)



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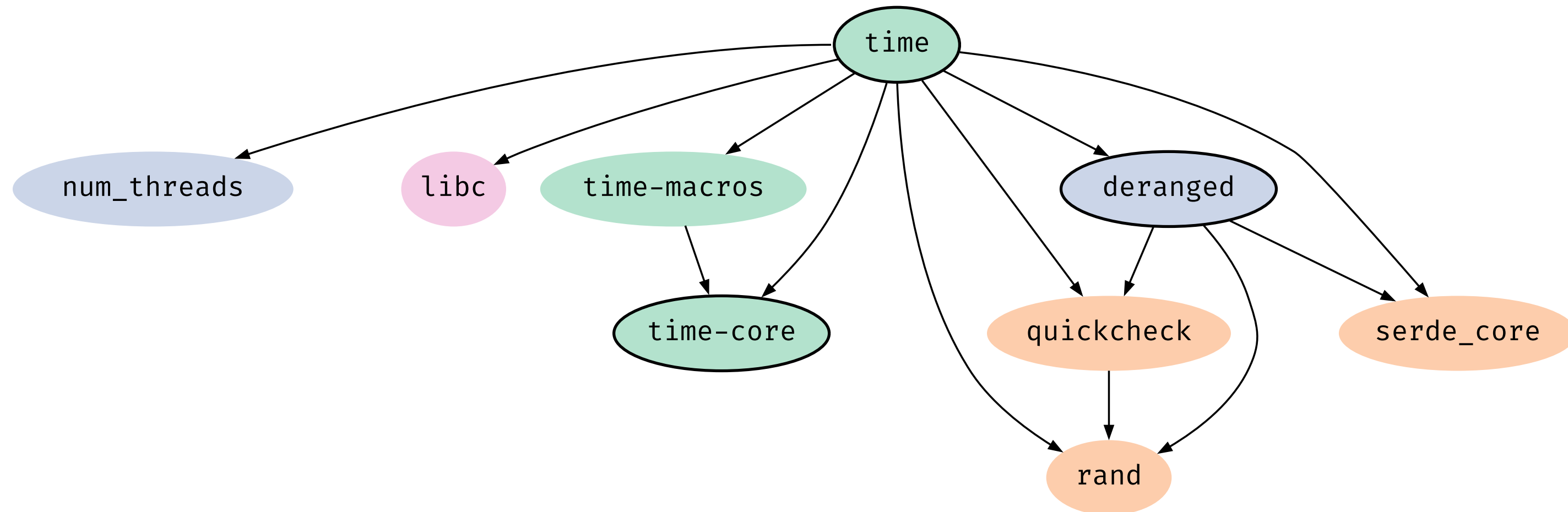
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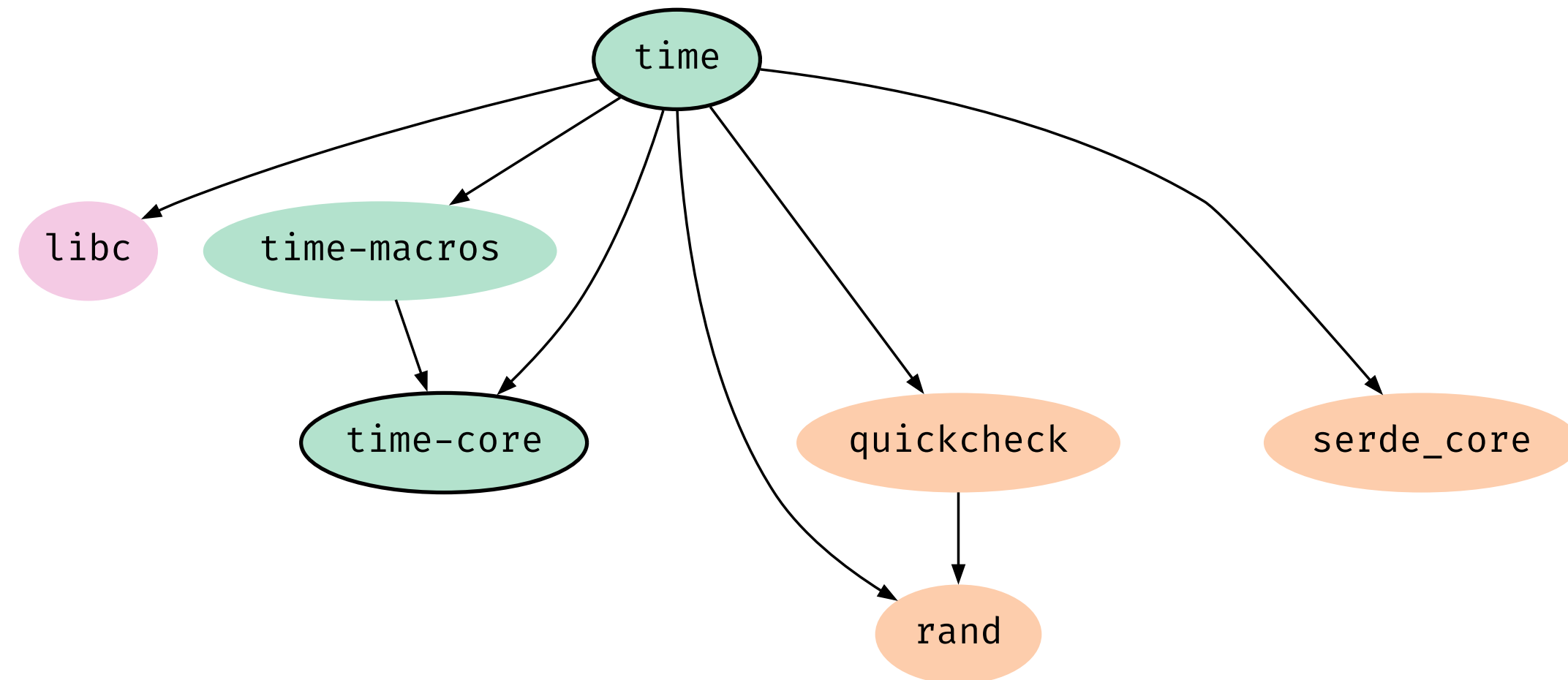
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Formatting and Parsing

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well-known	ISO 8601 (international standard) RFC 2822 (email) RFC 3339 (similar to ISO 8601)

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Format descriptions

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time	[weekday repr:short] [month repr:short] [day]
well-known	ISO 8601 (international standard) RFC 2822 (email) RFC 3339 (similar to ISO 8601)

Example of *completely valid* RFC 2822:

HT Sat, CR
(HT foo012F00!) (SOH VT SO DEL) (\ NUL) (\ SOH \ HT \ (\) \ \ \ DEL) \
(\
 \ VT) 02
2021 03:04:05 GMT

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let fmt_rt: Vec<_> = parse_owned::<2>(  
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const FMT_V2: &[BorrowedFormatItem<'_>] =
    format_description!(
        version = 2,
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const FMT_V3: FormatDescriptionV3<'_> =
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    );
```

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```

Pushing the Compiler

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- Simulating `const impl Default`

```
impl T {  
    pub const fn default() → Self { /* ... */ }  
}
```

Pushing the Compiler

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Pushing the Compiler

- Simulating const impl Default
- Eliminating magic constants

```
Second :: per_t :: <u32>(Week) // 604,800 as a constant
//
//           _____
//           named / anonymous
```

Pushing the Compiler

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Pushing the Compiler

- Simulating `const impl Default`
- Eliminating magic constants
- Type-level assertions

```
Time :: MIDNIGHT.format(&Iso8601 :: DATE) // compile error
```

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Pushing the Compiler

- Simulating `const impl Default`
- Eliminating magic constants
- Type-level assertions
- `const` generic ADTs

```
pub type EncodedConfig = u128; // explicitly unspecified
```

```
impl Config {  
    pub const fn encode(self) → EncodedConfig { /* ... */ }  
    pub const fn decode(_: u128) → Self { /* ... */ }  
}
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 - day of year $\rightarrow$ month/day conversion
 - 2.35 $\times$ throughput with new algorithm



Optimizing with
Novel Calendrical
Algorithms

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- similar story with date $\longleftrightarrow$ integer mapping
- Resources: *Baum, Hinnant, Neri, Schneider, Drepper, Joffe*



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Performance

(elapsed time, smaller is better)

Baseline



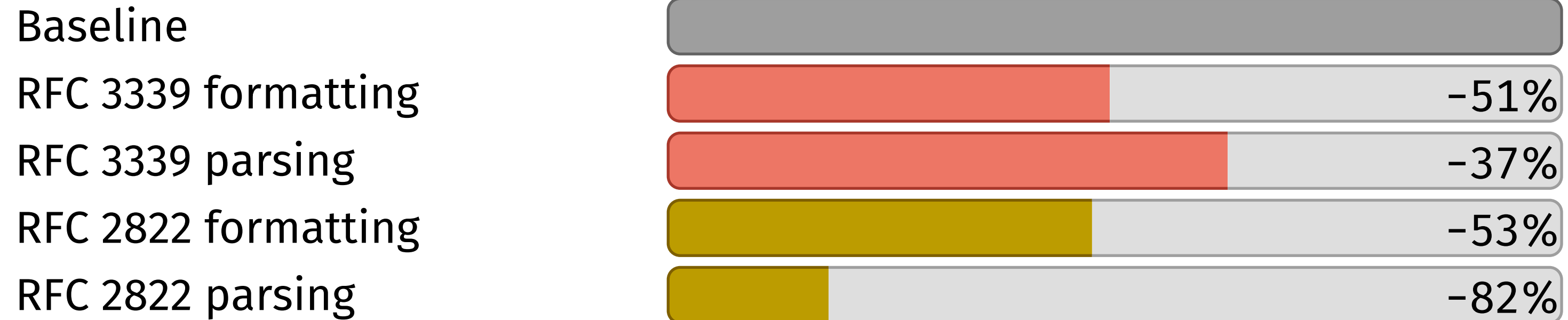
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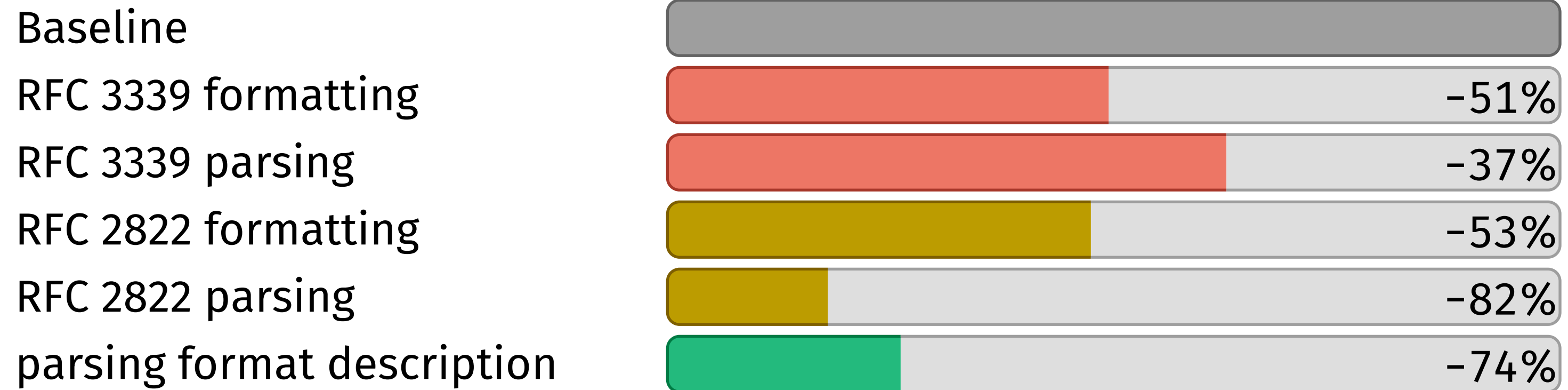
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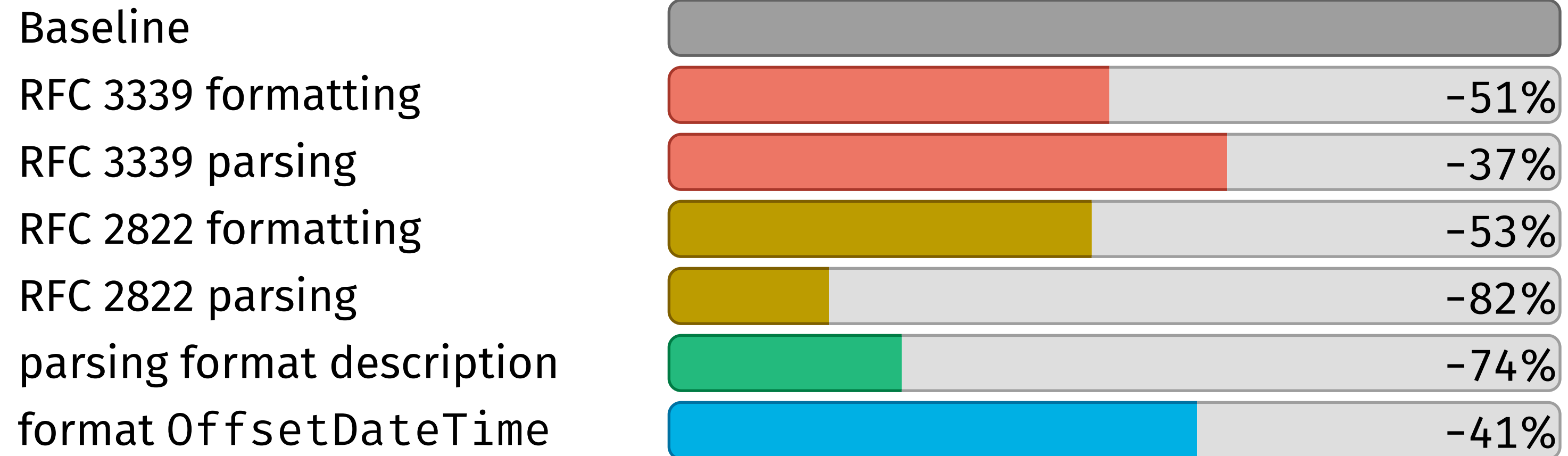
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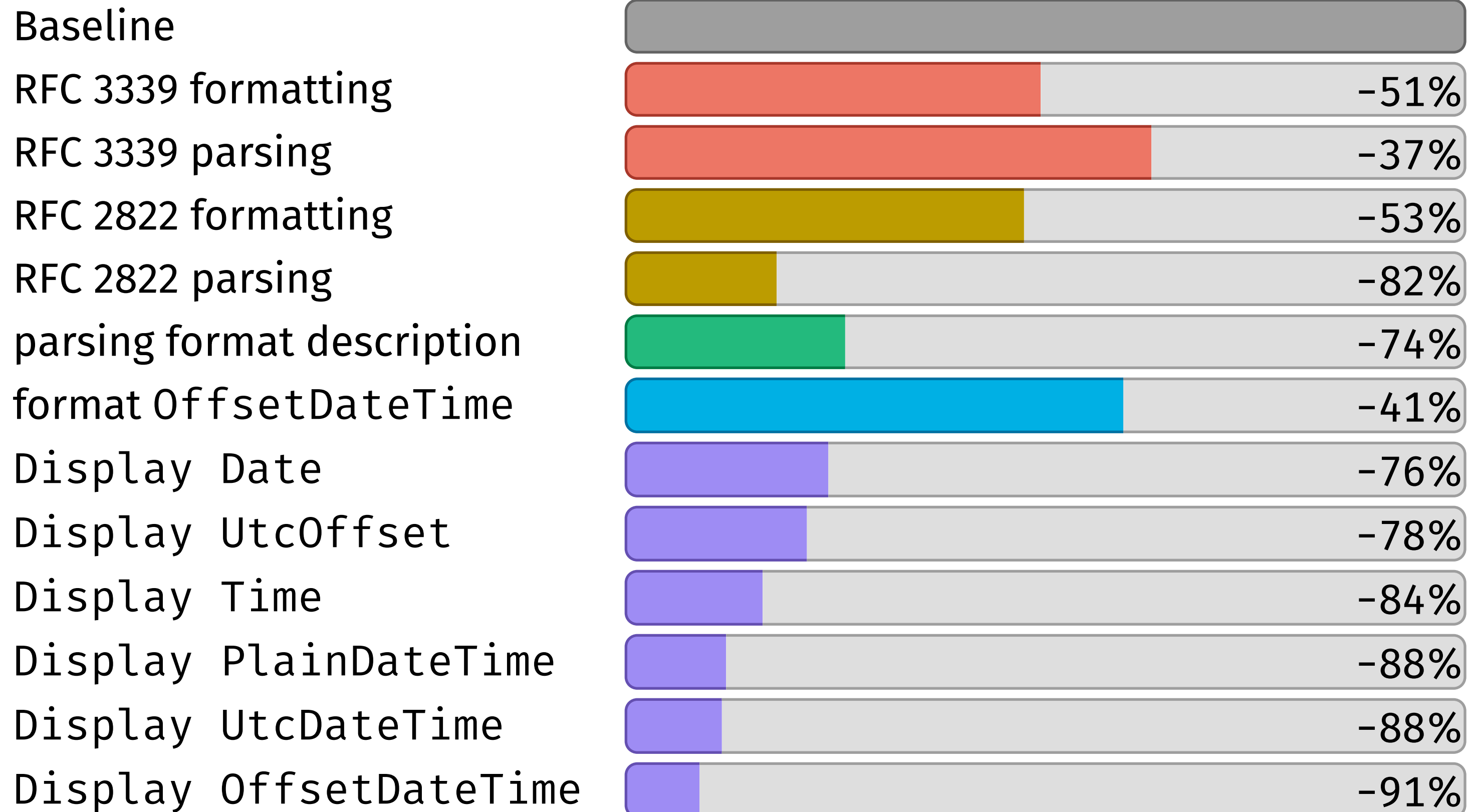
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#[derive(Clone, Copy)]
struct Digits {
    _padding: MaybeUninit<[u8; 7]>,
    digit_1: u8,
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let [
    digit_1,
    digits_2_and_3, // 6's static str
    // ...
] = subsecond_from_nanos(n);
let buf = Digits {
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#[repr(u8)]  
enum Padding {  
    Optimization,  
}
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CI/CD

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- sharding publishing reduced time by a full hour

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prepare miri

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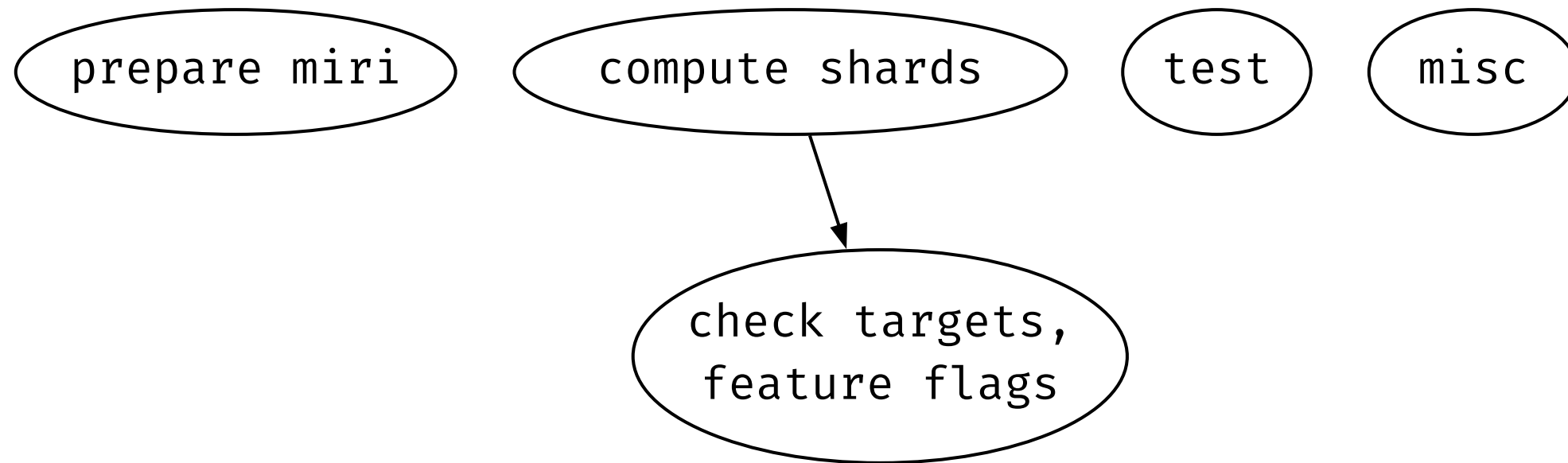
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misc

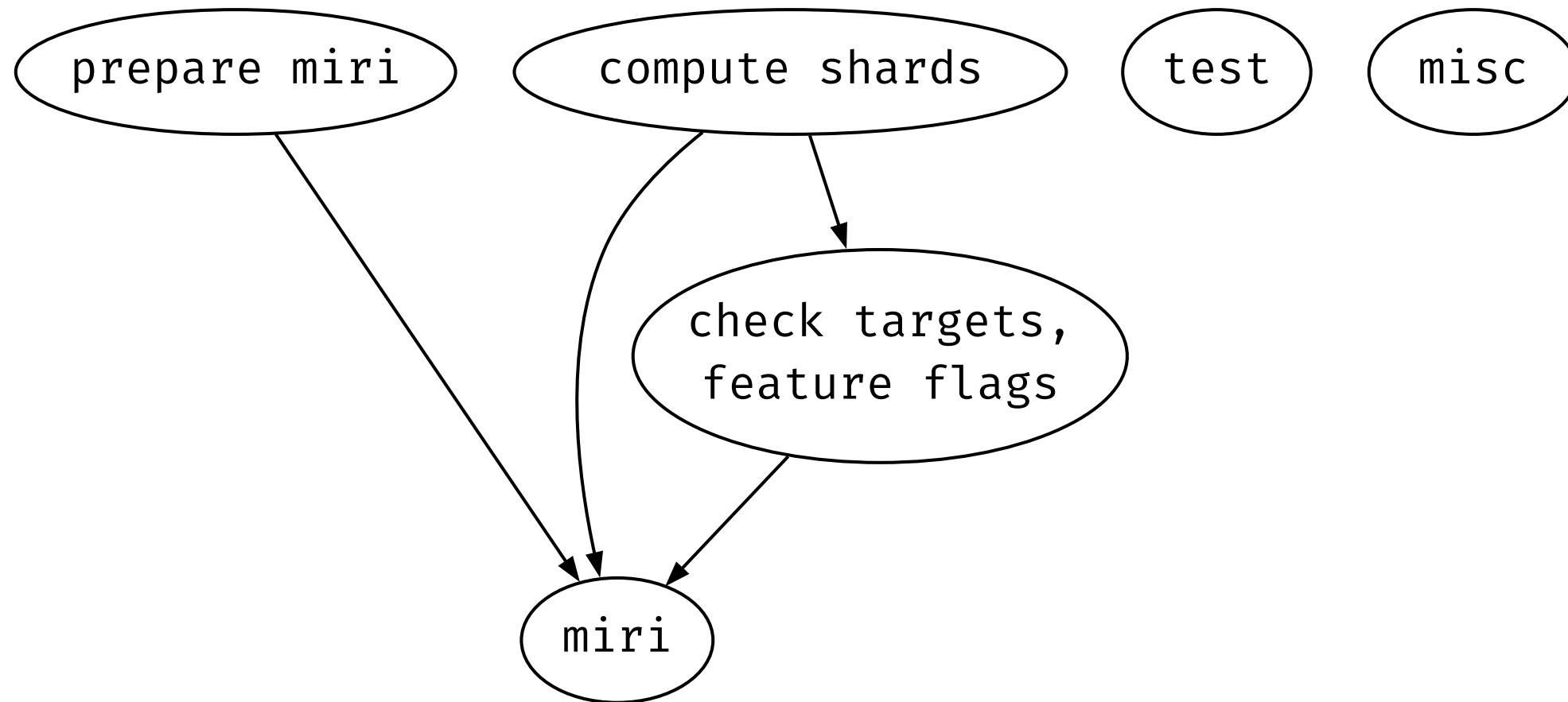
CI/CD



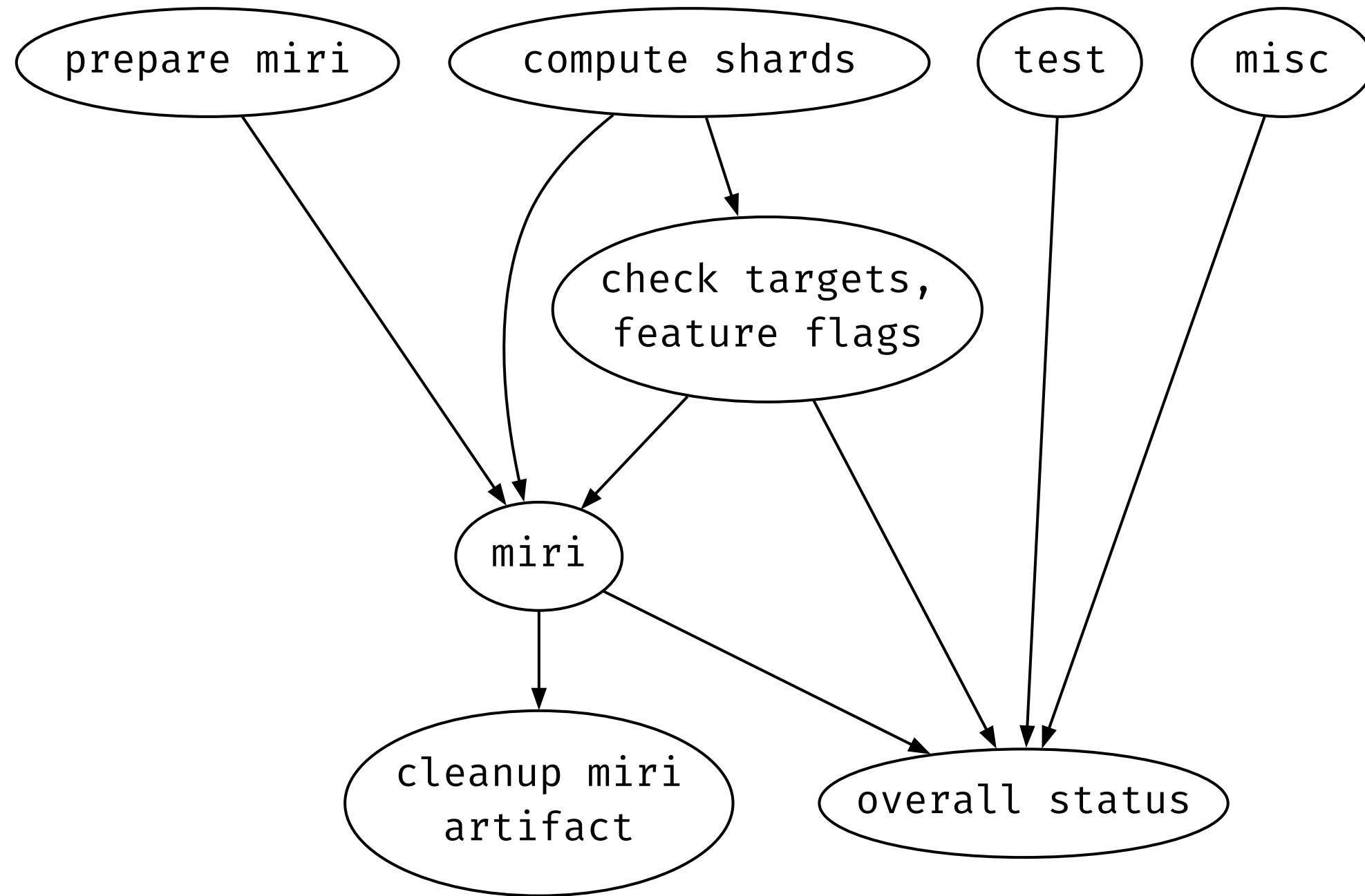
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 - `&&`, `||`, `!` in `cfg` — needs T-lang decision

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1.0 release

Tick Tock: Maintaining Time



Jacob Pratt



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2026-09-10
Montréal



RustConf
2026